Formulation of OFT We begin by defining the OFT as follows. Consider $\alpha = \sum_{i=1}^{n} \mathbf{a}_i \delta_{\mathbf{v}_i}$ and $\beta = \sum_{i=1}^{n} \mathbf{b}_i \delta_{\mathbf{v}_i}$ are two measures on the graph $G(\mathcal{V}, \mathcal{E})$, where $N = |\mathcal{V}|$ and $(\mathbf{a}, \mathbf{b})$ are two balanced vectors satisfying $< \mathbf{a} - \mathbf{b}, \mathbf{1} >= 0$. The formulation of OFT can be specified as:

$$\min_{\mathbf{P} \geq \mathbf{0}} < \mathbf{D}, \mathbf{P} > \quad \text{s.t.} \quad \mathbf{P}\mathbf{1}_N - \mathbf{P}^\top \mathbf{1}_N = \mathbf{s} \tag{2}$$

where $\mathbf{D} \geq \mathbf{0}$ is the distance matrix and $\mathbf{s} = \mathbf{a} - \mathbf{b}$. Note for each node $\mathbf{v}_i \in \mathcal{V}$, if $\mathbf{s}_i > 0$, we refer to node $\mathbf{v}_i$ as a supply node with a supply value of $\mathbf{s}_i > 0$. If $\mathbf{s}_i < 0$, it is a demand node with a demand of $-\mathbf{s}_i$. If $-\mathbf{s}_i = 0$, we categorize node $\mathbf{v}_i$ as a transshipment node. Besides, considering each edge $e_{ij} \in \mathcal{E}$, $\mathbf{D}_{ij}$ is the distance of the edge e_{ij} and we set $\mathbf{D}_{ii} \to +\infty$ to prevent self-transportation of nodes here. In comparison to traditional vanilla OT, our OFT considers flow balance constraints $\mathbf{P}\mathbf{1}_N - \mathbf{P}^\top \mathbf{1}_N = \mathbf{s}$ instead of the marginal constraints in $U(\mathbf{a}, \mathbf{b})$ in Eq. 1.

Exact Solving for OFT For solving the optimization in Eq. 2, in the field of OT, solutions are typically obtained indirectly. Specifically, one can first define the cost matrix using the geodesic distance (or shortest path metric):

$$\mathbf{C}_{ij} = \min_{K \geq 0, (i_k)_k : i \to j} \Big\{ \sum_{k=1}^{K-1} D_{i_k, i_{k+1}}, 0 \leq k \leq K, e_{i_k, i_{k+1} \in \mathcal{E}} \Big\} \tag{3}$$

where $i \to j$ indicates $i_1 = i$ and $i_K = j$, which is the path starts at i and ends at j. The primary formulation of OT, or called the 1-Wasserstein distance can be formulated as

$$\min_{\pi \geq \mathbf{0}} < \mathbf{C}, \pi > \quad \text{s.t.} \quad \pi \mathbf{1}_N = \mathbf{a}, \pi^\top \mathbf{1}_N = \mathbf{b} \tag{4}$$

where $\mathbf{pi}$ represents the final transportation results between the source and target nodes, which can

be equivalent to $\mathbf{P}$ given the routing of the shortest path in Eq. 3. Then one can adopt the Network Simplex (Dantzig, 1951) or Sinkhorn Algorithms (Cuturi, 2013) to get the solutions. However, these shortest path-based approachs have some notable limitations because the routes in the graph are fixed, and there is no way to impose constraints on the flow of any arbitrary edge or node, which limits the potential for real applications such as real traffic scenarios. Another line of approaches (Li et al., 2010; Mohammad Ebrahim & Razmi, 2009) is to use heuristic algorithms based on minimum cost flow. However, different from the algorithms of transportation problems, most of these algorithms assume that all data are integers (e.g. (de Vos, 2023)) and aim to find an integer-valued flow with the minimum total cost while meeting the supply-demand constraints for the graph. This severely limits the algorithms' applications on high-precision data.

3.2 Optimal Flow Transport with Entropic Regularization

In this subsection, we propose the definition of entropic OFT and corresponding OFT-Sinkhorn to get the approximate solution for OFT.

Formulation for Entropic OFT Differing from previous CPU-based algorithms, in this paper, following (Cuturi, 2013), we consider the entropic OFT and employ a GPU-friendly matrix-vector iterative algorithm to speed up the computations for OFT. However, directly adding entropy regularization cannot derive an algorithm to obtain an approximate solution as directly as vanilla OT. The reason is that: 1) there are many isolated points that do not have any flow passing through them in the exact solution, while under entropy regularization, flows are necessarily channeled through them, causing significant bias; 2) the flow balance constraint cannot directly form an alternating iterative algorithm like

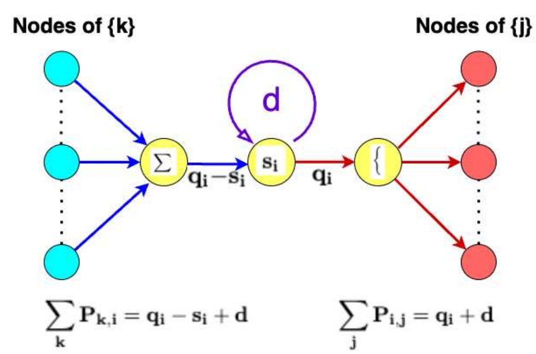

Figure 2: **Illustration of our constraints in Entropic OFT.** The flow constraints satisfied by Eq. 5, where each node includes virtual flow($\mathbf{d}$), input flow($\mathbf{q_i}$), and output flow($\mathbf{q_i} - \mathbf{s_i} + \mathbf{d}$).